Praise Worthy Prize

International Review on Modelling and Simulations (I.RE.MO.S.), Vol. 14, N. 1
ISSN 1974-9821 *February 2021*

Isolated Microgrid Management Using a Multi-Agent System

B. Boudiaf, S. Zebirate, N. Aissani, A. Chaker

Abstract – *Solar energy offers a considerable potential for the energy market and it brings many challenges for the actual research. In order to meet the economic and the environmental expectation, the management of an isolated microgrid needs to be more efficient and profitable. In this paper a Multi-Agent System (MAS) is proposed in order to manage an isolated photovoltaic microgrid. The proposed approach presented in this paper improves the management of an isolated decentralized microgrid. It takes in account load voltage constraints, ensures the power demand, reduces chattering phenomenon on the storage system, and it maintains the photovoltaic panel operating in its maximum efficiency point. Finally, the proposed method is evaluated throw three scenarios, and the results are obtained by simulation using MATLAB/Simulink and JADE (Java Agent Development Environment).* ***Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved.***

Keywords: *Photovoltaic System, Multi-Agent System, Autonomous Microgrid, MATLAB/Simulink, JADE*

Nomenclature

ACK	Acknowledgement
ACL	Agent Communication Language
Battery_agent	Battery agent id in the MAS
Behavior	Classes representative agent tasks
Com_agent_a	First communication agent id in the MAS
Com_agent_b	Second communication agent id in the MAS
Cyclic behavior	A behavior that must be executed forever
e_S	Static error
JADE	Java Agent Development Environment
Load_agent	Load agent id in the MAS
MAS	Multi-Agent System
MPPT	Maximum Power Point Tracking
NAK	Negative acknowledgement
P_{Bat}	Battery power
P_{Load}	Load power
$P_{\max}$	Maximum power witch the photovoltaic system can deliver in the optimal condition
PPV	Photovoltaic panel power
PSO	Particle Swarm Optimization
PV	Photovoltaic
Pv_agent	Photovoltaic agent id in the MAS
r	Line resistance
TCP/IP	Transmission Control Protocol/Internet Protocol
UPQC	Unified Power Quality Conditioner
V_{Bat}	Battery voltage
V_{Load}	Load voltage
V_{PV}	Photovoltaic panel voltage

I. Introduction

Around the world, renewable energy use is on the rise and these alternative energy sources could hold the key to fight climate changes. Solar systems have a large potential and take a significant place on the actual research. In order to meet the challenge, it is needed to develop more efficient and economic systems, adapting the infrastructures, integrate conversion and storage systems. In order to ensure the demand, optimize the economic criteria, and increase the equipment lifetime, several strategies are addressed in the literature. A short and long-term storage system is used in an algorithm-based approach. Fuel cells and electrolyzes are used for long-term storage; they ensure the energy demand when the photovoltaic panel cannot. The batteries are used as a short-term storage system to stabilize the power balance.

This increases the lifetime of the hydrogen storage system. However, it increases the use of the battery [1].

A modified Particle Swarm Optimization (PSO) is used in [2] to insure the photovoltaic panel MPPT system. It is used to optimize the energy transfer efficiency of the battery charging operation. In [3], the author uses a mixed-integer linear optimization algorithm in order to optimize the photovoltaic system power production, and the use of the storage system. The algorithm uses forecast data and calculates the round-trip efficiency of the storage system with the objective of maximizing the photovoltaic production. Then the method is then applied to an existing low voltage microgrid. In [4], a robust multi-stage method is proposed. The objective is to solve operational problems for the presented isolated system, using a cutting plane algorithm and bender's decomposition through a pumped hydroelectric storage system. The results presented in the

Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved https://doi.org/10.15866/iremos.v14i1.18940

paper demonstrate the efficiency and the adaptability of a multi-stages strategy against power fluctuations. Other approaches focus on the system's performance rather than on lifetime. An intelligence approach is used on [5], where an artificial neural network is used to optimize the photovoltaic MPPT system. The results are validated by a comparison with a P&O (Perturb and Observe) algorithm. An algorithm-based approach aims to improve the electrolysis process. The strategy uses conditions such as a minimum power to start the electrolyzation.

This constraint improves the hydrogen system's performance and lifetime but it involves a significant use of the battery, especially on low power balance. The strategy in [6]-[8] uses different approaches to solve the optimization problem in order to maximize hydrogen production. In [9], a fuzzy logic controller is used to improve a Unified Power Quality Conditioner (UPQC) operation. The photovoltaic panel and a battery storage system are connected to a DC. The controller maintains the bus voltage from the disturbances. In [10], a sliding mode approach is used for the photovoltaic system maximum power point optimization. The process is a pumping system based on a DC motor coupled with a centrifugal pump. The results are done by simulation using MATLAB/Simulink. In [11], a Neural Network MPPT based is used on an isolated photovoltaic system.

The process considered is a 1.8 kW led lamp park using connected in a DC bus. The controller acts on the buck-boost converter PWM that powers the LEDs. Additional approaches are used, based on the economic criteria. In order to get the optimal solution, different cost functions assigned to each equipment are calculated. The optimal solution for cost reduction is found using Model Predictive Control [12], Flow Chart & Genetic Algorithm [13], Fuzzy logic & Differential Evolution Algorithm [14]. A robust two- stage optimization strategy is applied to a grid connected system in [15].

The objective is to minimize the long-term storage system operational cost by taking in account the storage system proper use constraints. The first optimization stage schedules conventional generators hourly unit commitments, while the second stage performs real time scheduling for the energy trading, energy storage system and conventional generators economic dispatch. In [16], a smart energy management system is developed in order to optimize an isolated microgrid operation. The microgrid is designed using an Affine Arithmetic method. The method is used to compute the affine forms of the noise symbols values in order to obtain a cost-effective solution on real time power dispatch. The paper considers the use of the storage system to estimate the energy price and it determines the optimal operation strategy. In [17], a quadratic programming method is used for an optimal power dispatch of an interconnected microgrid. The paper presents a centralized approach and it introduces a concept of end user driven microgrid. This gives more flexibility to users to manage the battery and to minimize the power expected from the grid. The optimization is based on economic criteria with the constraint of satisfy the demand. The approach has been tested on a modified IEEE 33 bus distribution feeder. In [18], the paper treats the problem of day-ahead scheduling of generation and storage in a microgrid. The objective is to minimize the cost of the produced energy.

The load and the renewable sources are modeled using a proven uncertainty model, and the optimization is done via an artificial fish swarm intelligence algorithm. The microgrid includes a diesel generator, a fuel cell, a micro-turbine, two photovoltaic generators, and a wind turbine. In [19], the paper intends to optimize microgrid sizing in order to minimize the total system cost. It includes a photovoltaic system, a biomass gasifier, and a diesel engine. The control problem is formulated as a Markovian decision process, and the optimal solution is found by reinforcement learning algorithm. In [20], a reinforcement learning method is applied for an optimal microgrid control. The system includes renewable sources and batteries storage system. In the paper, several reinforcement learning agents have been developed, and then they have been compared to each other and to a mixed-integer linear programming method. The objective is to minimize the total installation cost. The last approach includes the technical-economic criteria. It aims to minimize the maintenance and production cost and to maximize the equipment life. The optimization problem is nonlinear with a multi-objective function. A Fuzzy Logic, Particle Swarm Optimization Algorithm, and Linear Programming approach are used in [21]-[23].

In [24], a flow chart is proposed in order to ensure the technical and the economic objectives. A particle swarm algorithm for real-time energy management is presented in [25]. The method is applied on a grid-connected microgrid with the objectives of minimizing the waste of the energy and production cost. Several scenarios have been analyzed to examine the suitability of the cost function presented, and the results show that the method presented has reduced the costs by 12% over a period of 96 hours. In [26], a technical and economic approach is applied on an isolated microgrid. The paper aims to ensure the robustness and cost-effectiveness of the microgrid by simulation. The system is modeled using HOMER Pro software and it is optimized using a mixed integer linear programming method, where cost function is calculated by a general algebraic modeling system. In [27], a comprehensive stochastic energy management system is applied to a hybrid microgrid. The system represents a day-ahead management that improves the reliability of the microgrid and minimizes the operational cost. Thermal and electrical load are considered in the paper, as well as grid connected and off grid mode. A multi-objective optimization problem is posed and solved by particle swarm optimization and multi-objective particle swarm optimization algorithms. Results are done by simulation. Although many works have accomplished photovoltaic systems and isolated microgrids, it still presents several operational challenges. The interactions between power sources create local oscillation, and voltage control may change the photovoltaic system

Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved

operating point, and/or generate chattering phenomenon on behalf of the storage system. The approach chosen in this paper is using a Multi-Agent System (MAS) in order to manage a solar isolated microgrid installation. Multi-Agent Systems (MAS) are issued from the distributed artificial intelligence research. It integrates the cooperation between the different agents and human machine cooperation as well. The use of a MAS allows extending the actual researches on the energy system management, while introducing a different method to solve an isolated microgrid power management problem.

This paper is organized as follows. In Section II, the problem statement treated in this work is described.

Then, in Section III, the system architecture is introduced. Afterwards, the process and the Multi-Agent system are developed in Sections IV and V. Next, in Section VI, the suitability of the presented method is verified by simulation using MATLAB/Simulink and the JAVA platform. After that, in Section VII, the discussion of the simulation results is presented. Finally, Section VIII includes the conclusion of this work.

II. Problem Statement

Each one of the strategies cited above offers its own advantages. However, implementing such a system still involves considering load side constraints. In an isolated system topology, the difference between the produced power and the demand leads to high switching frequency between the photovoltaic panel and the battery [28]. This causes an excessive use of the storage system, and it implies to change the operating point of the photovoltaic panel, which reduces the MPPT system efficiency. On another hand, when the load is powered by the photovoltaic system and the battery, the two-power sources are set to be in parallel (Fig. 1) in order to comply with the DC Bus voltage (or the AC Bus voltage, depending on the system topology). According to (3), this will prevent the photovoltaic panel to operate at its maximum efficiency point:

$$P_{Load} = V_{Bat} + ri_1 + V_{PV} + ri_2 \quad (1)$$

$$V_{Bat} + ri_1 = V_{PV} + ri_2 = V_{Load} \quad (2)$$

$$P_{Bat} = P_{PV} \quad (3)$$

where V_{Load} is the load voltage, V_{Bat} is the battery voltage, V_{PV} is the Photovoltaic panel voltage, r is the line resistance

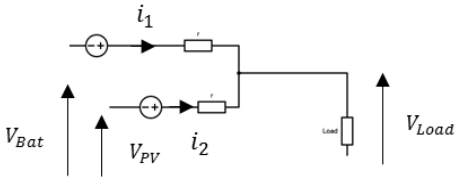


Fig. 1. Wind turbine power versus the DC bus voltage at different rotor speeds

The isolated micro-grid considered in this paper is completely decentralized. It is composed of a photovoltaic system and a battery storage. A multi-agent bases management strategy is proposed in order to ensure the demand, maintain the load voltage, and ensure the photovoltaic panel to operate always on its maximum efficiency point. It takes into account the proper use of the battery and eliminates the high switching frequency chattering mode.

III. System Structure

A block diagram of the system is represented on Fig. 2. It includes the process, the Multi-Agent System (MAS) and a supervisor that connects them. The process and the supervisor are modeled under Matlab/Simulink and the MAS is developed under JAVA using JADE platform (Java Agent Development Environment). The process shown in Fig. 3 includes a photovoltaic panel, a battery, and a load. The battery stores the extra energy produced by the photovoltaic system, or it switches to source mode when the power demand is superior to the photovoltaic system's production. The system aims to be decentralized and modulable, where the source and the load can be easily added or removed. This involves choosing an AC bus to connect its components. The supervisor (shown in Fig. 4) ensures the connection between Simulink and the MAS. When it starts, it initializes the connection with the MAS. Then, it collects continuously data from the process to send them to the MAS via TCP/IP. Finally, the supervisor receives the setpoints of the electrical converters and the switches from the MAS, and relays them to the process.

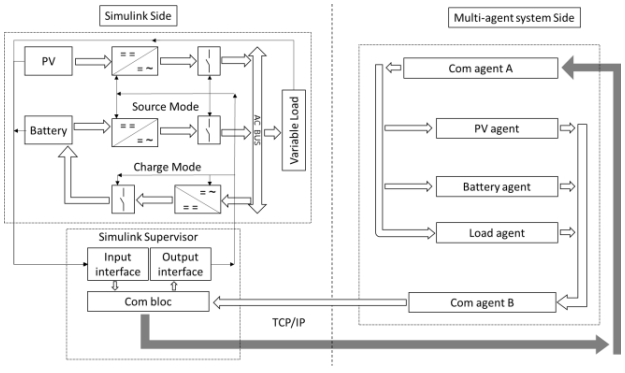


Fig. 2. Micro-grid system structure

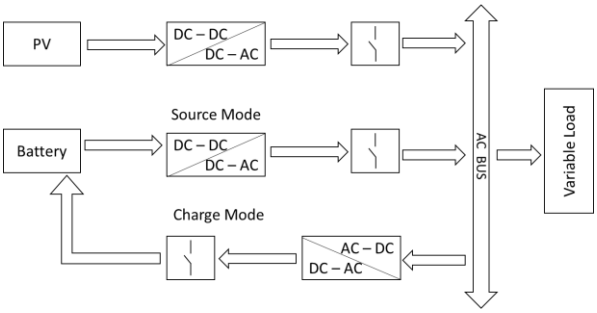


Fig. 3. Simulink model

Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved

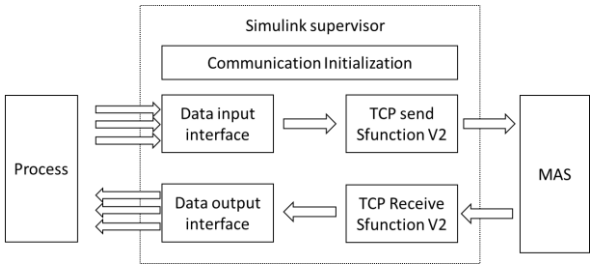


Fig. 4. Simulink supervisor

The MAS controls the output of the batteries and the PV (Photovoltaic) electrical converters and switches. It is composed of five agents: the three agents, <<pv_agent>>, <<battery_agent>> and <<load_agent>>, use a continuous negotiation process to manage the electrical production, while the two other agents, << com_agent_a >> and << com_agent_b >>, manage the communication between the MAS and Matlab/Simulink.

The MAS communication platform allows agents to communicate with each other. They use ACL messages (Agent Communication Language) during the negotiation process.

IV. Simulink System Structure

The Simulink system is developed by the process model and the supervisor, which are shown in Fig. 5. The photovoltaic panel is connected to the bus through a DC-DC Boost converter, then to a DC-AC inverter and an electrical switch. The battery charges from the bus through the AC-DC and DC-DC converters, and discharges through a boost DC-DC, then a DC-AC inverter. The supervisor detailed in Fig. 4 is composed of a communication initialization block, input/output interfaces, a TCP/IP send block and a TCP/IP receive block.

The connection between the process and the supervisor is shown in Figs. 6. The "Communication Initialization" block is a level 2 MATLAB S-function. It establishes a connection with the MAS when the simulation starts. The "Data input interface" collects all the process variables, attaches an ID to each variable in order to comply with the format <<xxxyyyzzz12345>> (Appendix) and sends them to the "TCP send" block.

The "TCP send" block sends the formatted data to the MAS via TCP/IP and it waits for an Acknowledgement (ACK) to send the next message.

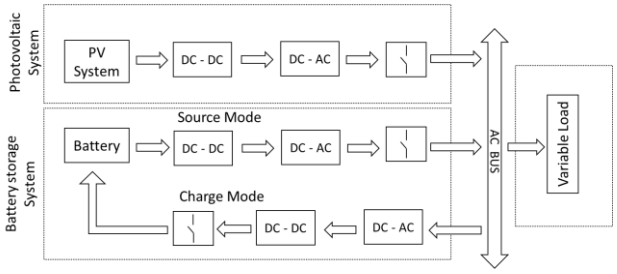
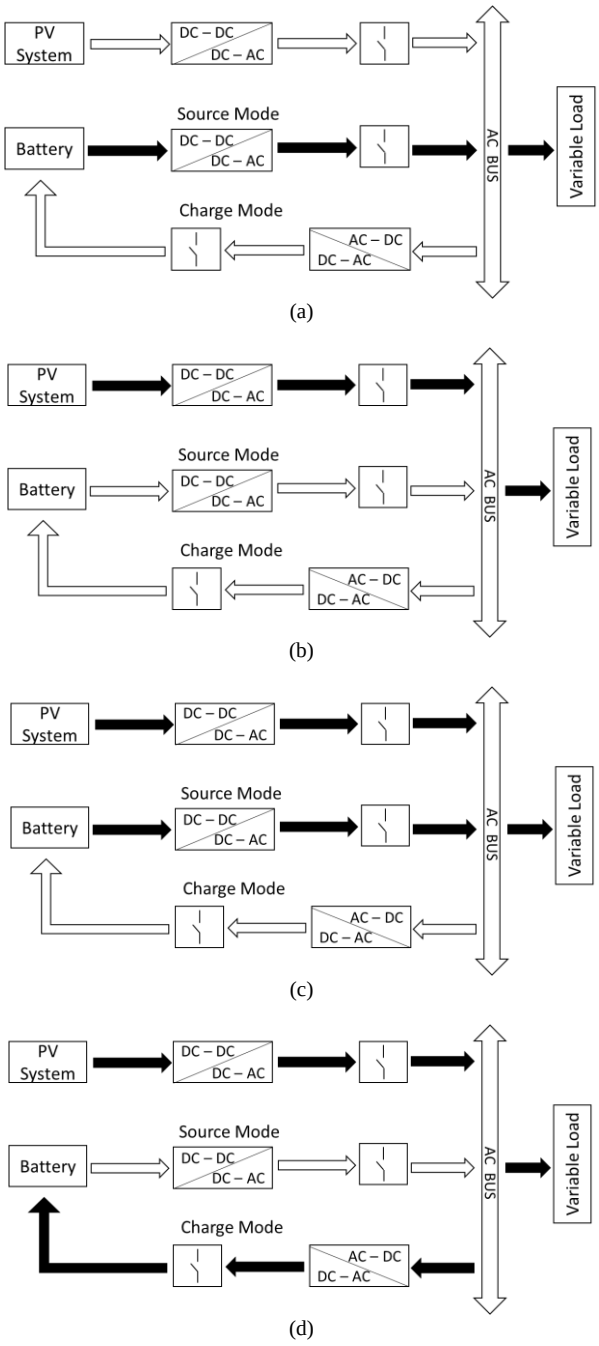


Fig. 5. Process scheme



Figs. 6. Process configurations:
(a), case 1; (b), case 2; (c), case 3; (d), case 4

If it receives a Negative-Acknowledgment (NAK), it sends the same message again. The "TCP Receive" block receives messages from the MAS. When the message complies with the format in use (Appendix) it replies with an ACK and relays the message to the "Data output interface", otherwise it replies with a NAK. The "Data output interface" updates its output according to the data received from the "TCP Receive" block.

V. Multi-Agent System Structure

In the presented MAS, the communication system is

Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved

designed by two agents and the management system by the three others. Each agent is described of one or more behaviors (Behaviors are classes that contain the code to be executed by the agent after the initialization phase).

V.1. Communication Platform Structure

This structure ensures the connection with MATLAB/Simulink. It is performed by the "Com Agent A" and the "Com Agent B". The "Com agent A" aims to receive messages from MATLAB/Simulink. It creates a socket server and waits for data. When the agent receives a message, respecting the format used (Appendix), it reruns with an ACK and relays it to the agent recipient.

Otherwise, it reruns with a NAK. The "Com agent B" aims to relay messages from the other agents to MATLAB/Simulink. It creates a second socket server and waits for MATLAB/Simulink to initialize the connection before transmitting agent messages to it. When it receives back an ACK, it sends the next message; otherwise, it resends it.

V.2. The Management System Structure

The management system is shown on Fig. 7. It is composed of the "Battery agent", the "load agent", and the "pv agent". These three agents operate a negotiation process in order to achieve the objectives cited above.

This process uses the concept of a market between energy producer and consumers, where the consumer queries the producers and negotiates for the best deal.

The load agent queries the electricity producers for their offers. Based on the information received, it establishes a power profile and submits the proposal to each agent. On the other side, the power sources agents consider their equipment's proper use constraints. This process leads to one of the four cases shown respectively in Figures 6(a), 6(b), 6(c), and 6(d). In the first case, the battery is in source mode and the photovoltaic panel is disconnected. This happens when there is no solar radiation and the battery's state of charge is not below the minimum safety state of charge. In the second case, the photovoltaic panel is connected while the battery is disconnected.

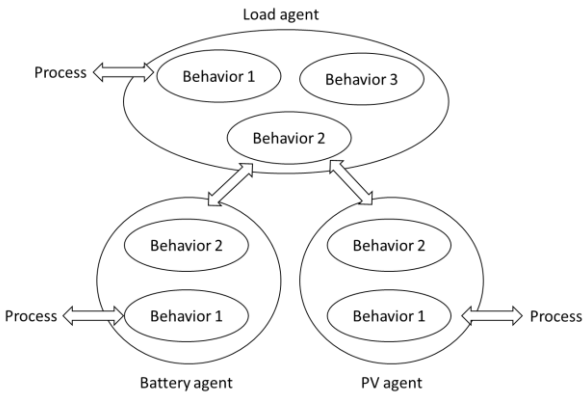


Fig. 7. Management system structure

This happens when the photovoltaic panel power meets the load demands, or when the battery is fully charged, and the photovoltaic panel satisfies the demand. In the third case, both the photovoltaic panel and the battery power the load. This happens when the load demand is superior to the photovoltaic panel production.

In the fourth case, the photovoltaic panel is connected, and the battery is in charge mode. This happens when the power demand is inferior to photovoltaic production.

The "battery agent" is formed by two cyclic behaviors.

The first one receives data from the storage system and sends back the switches and converters setpoint. These setpoints are function results from the negotiation process and batteries-appropriate usage parameters. The second behavior manages the negotiation. The "PV agent" is composed of two cyclic behaviors. The first one receives the data from the photovoltaic system and controls the switch and the converter output. The second behavior aims to handle the negotiations with the other agents. The "load agent" supervises the negotiation. It is characterized by three behaviors. The first one is a cyclic behavior. It receives data from MATLAB/Simulink in order to keep the load profile updated. The second one is a cyclic behavior. It establishes a four steps negotiation with the other agents. Each step is executed until specific conditions are met.

- *Step 1*: The first step consists to request data from each power source process, and the maximum power that the source can deliver;
- *Step 2:*In this phase, the agent waits for the Battery and PV agents reruns;
- *Step 3:*In this phase, the agent uses the load, the battery, and photovoltaic data to establish a power profile and set the percentage of power needed from each source. Next, it calculates the voltage output of the PV, and battery's converters in order to meet the load voltage criteria with the power profile established. Finally, it submits the proposal to the other agents;
- *Step 4:*In this phase, the agent waits for the proposal replay. Once it receives all the replays, negotiation can start over from the first step.

The third behavior uses a Ticker of 100 ms to reinitialize the negotiations process. This allows providing a timeout to limit the time spent waiting for an answer. The agents sequence diagram shown in Fig. 8 illustrates the communication process. Communication agent "Com Agent A" receives process data to update every agent's process variables ("PV agent" and "battery agent").

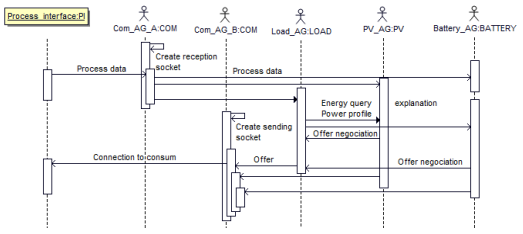


Fig. 8. Agents communication sequence diagram

Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved

"Load agent" receives load process data and initiates a negotiation process with the other agents. Based on their offers, it establishes a power profile and performs a proposal. At the end of the negotiation round, each one of these three agents sends control variables to the process via the communication agent B "Com Agent A".

VI. Simulation Results

In order to analyze the performances of the proposed method, several simulations have been done in MATLAB/Simulink and Eclipse/Java. The parameters used are listed in Table I. It has been chosen to present the results of three scenarios, which are as follows.

VI.1. Linear Pursuit

In this scenario, linear variations are used for the photovoltaic power profile where the load is variable.

This scenario seeks to demonstrate the capability of the developed method to track maximum photovoltaic power, control the load output, response time, and battery's chattering removal. Simulation results are shown in Figs. 9-11. Photovoltaic power can be observed in Fig. 9 next to its reference. Fig. 10 illustrates the load voltage next to its reference, and Fig. 11 represents the battery charge/discharge signal.

TABLE I
SPECIFICATIONS ADOPTED FOR THE SIMULATION

Components	Variable name	Rating values
Battery	V_n	48 V
	Rated capacity	1.65 Ah
Load	P_{max}	145 VA-925 VA
	V_{eff}	220 V
	f	50 Hz
Line resistance	r	5 Ω

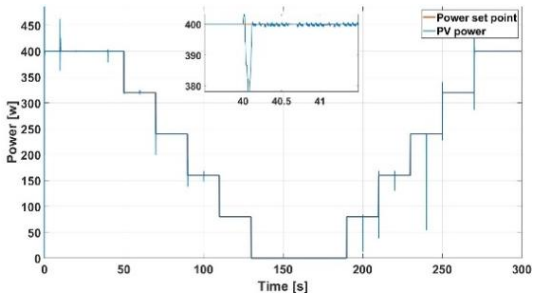


Fig. 9. Photovoltaic Power with its reference

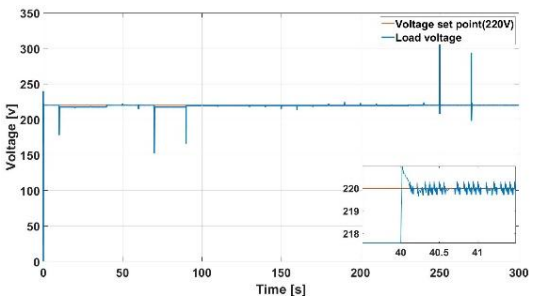


Fig. 10. Load voltage with its reference

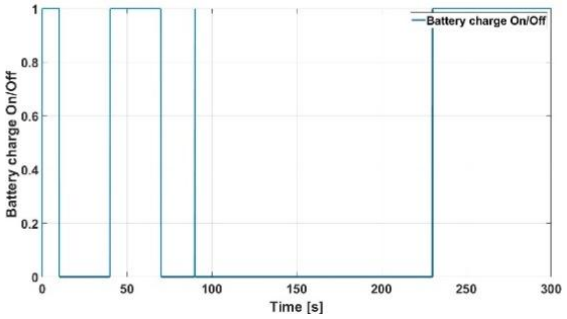


Fig. 11. Battery Charging Switch

VI.2. Second-Order Pursuit

In this scenario, Equation (4) is used for the photovoltaic power profile. The objective of this scenario is to demonstrate the management system's performance on non-linear PV power variation. This includes static error e_S, response time, of the photovoltaic power and load voltage, and battery chattering removal. Results are illustrated on Figs. 12-14. Fig. 12 shows the photovoltaic power next to its reference, the load voltage is shown on Fig. 13 next to its reference, and the battery charge/discharge signal is shown in Fig. 14:

$$P_{max} = -(t-15)^2 + t + 300 \quad (4)$$

VI.3. Solar Variation Impact

In this scenario, a photovoltaic power profile representing a solar cycle, including cloud perturbation has been used.

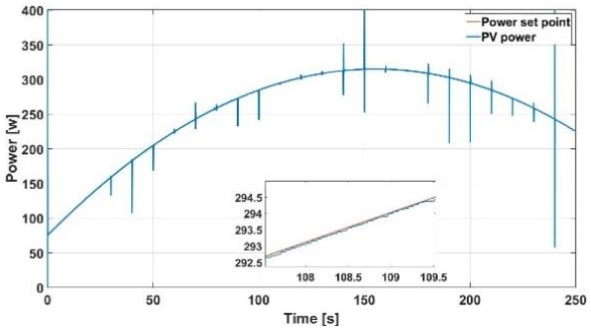


Fig. 12. Photovoltaic Power with its reference

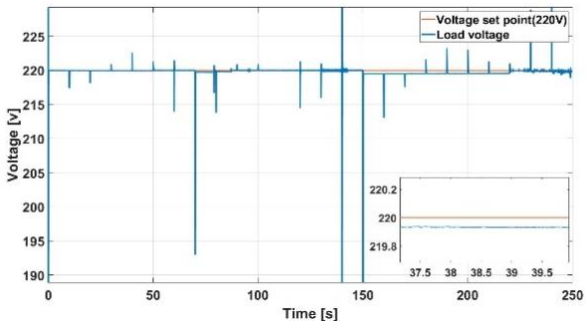


Fig. 13. Load voltage with its reference

Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved

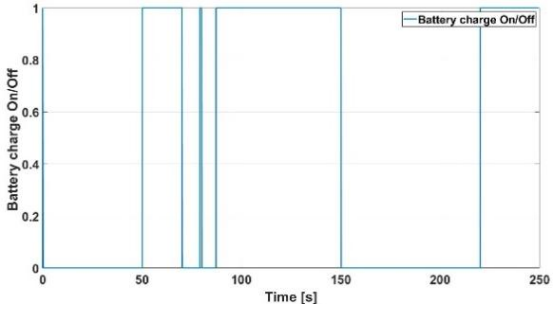


Fig. 14. Battery Charging Switch

The purpose of this test is to evaluate the influence of the solar variation on the PV system's operating power and the load voltage, the management system response time, and to verify its efficiency for battery's chattering removal. Simulation results are presented in Figs. 15-17. Solar variation is presented next to its reference in Fig. 15, load voltage is illustrated next to its reference in Fig. 16 and battery charge/discharge signal is illustrated in Fig. 17.

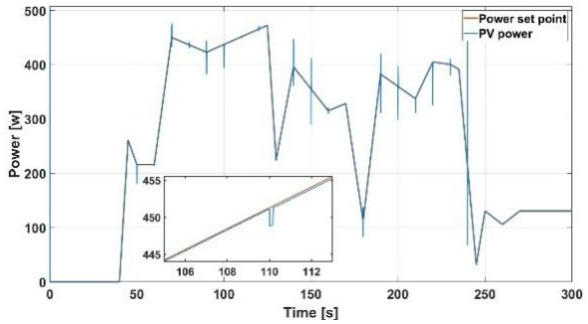


Fig. 15. Photovoltaic Power with its reference

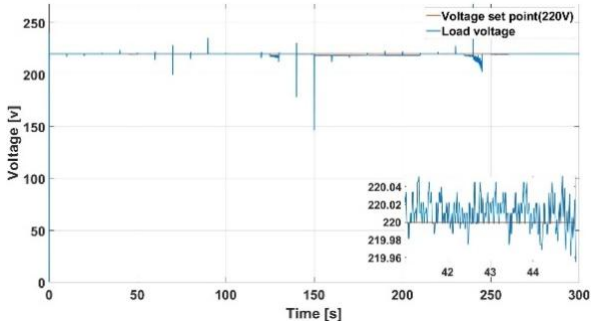


Fig. 16. Load voltage with its reference

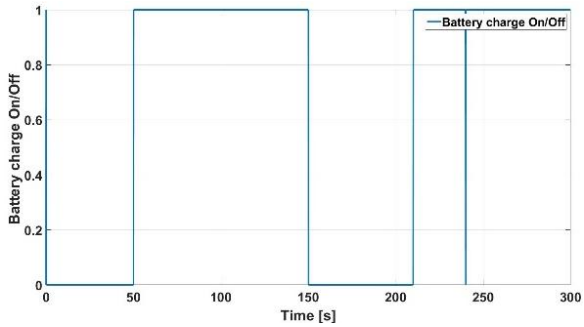


Fig. 17. Battery Charging Switch

VII. Discussion

The first scenario aims to validate the management system response to the PV power as well as load variations. The PV power and the load voltage are compared respectively to their references, and the battery charging/discharging switch signal is observed. At the beginning of the simulation, the PV system produces more power than required by the load. This allows the battery to be on charging mode. However, the increase of the demand allows the battery to power the load at the same time with the PV before it is switched back to charge. As the simulation goes, the PV power decreases to null, and it increases until the maximum again.

Meanwhile the load is powered with the PV and the battery, then it is powered with the battery alone, get back powered by the PV and battery and finally, the PV powers the load and charging the battery. In Fig. 9, the PV power next to its setpoint can be observed. The latter is formed of steps decreasing from the 400 W to 0 W, and then increasing back to 400 W. It is observed that the PV power tracks properly its setpoint; it is shown that the management system reacts very quickly to power variations as well as for load variations. Fig. 10 illustrates the load voltage next to its setpoint; the management system reaction is as fast as the PV power one. A static error can be observed where e_S stays inferior to 1%. Fig. 11 represents the battery charge/discharge signal. This shows the absence of the chattering on behalf of the storage system. In the second scenario, the PV maximum power is formed of a second order function.

This scenario aims to evaluate the management system response to nonlinear variation. The PV power tracking, the load voltage controls, the response time, and the capability of the developed management system of chattering suppression on behalf of the battery are examined. At the beginning of the simulation, the PV power is lower than the demand; the load is power with the PV and the battery at the same time. Since the load is variable, it allows the battery to switch to charging mode twice before the PV power reaches the maximum value.

After 150 s, the load is powered by the PV and the battery at the same time until t=220 s, where the load decreases. Then the PV powers the load and charges the battery at the same time. The PV power is illustrated next to its setpoint in Fig. 12. It is observed that the power tracks its reference with an acceptable static error, where e_S stays inferior to 1%. Fig. 13 illustrates the load voltage next to its reference. It is noticed that the static error can be slightly more important when the battery is on source mode but it stays always inferior to 1%. The response time for the voltage control is as fast as for the PV power tracking inferior where it stays inferior to 0.2 s. Fig. 14 illustrates the battery charge/discharge signal. This shows the absence of the chattering in the storage system controls. The third scenario aims to validate the developed method. The PV power profile describes a solar cycle including cloud perturbations. PV power tracking, and load voltage controls are analyzed. Static error and response time are verified as well as the

Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved

chattering suppression capability of the management system. The photovoltaic power is shown in Fig. 15 next to its setpoint. It can be attested that the developed approach had no difficulties to own the system at its maximum performance by charging as much as possible of energy instantly on-demand. Static error e_S is verified where it stays inferior to 1%. In Fig. 16, the load voltage next to its reference can be observed. It is observed that the load voltage tracks its reference with a static error e_S inferior to 1%. Response time is also verified for both PV power and load voltage. The latter remains lower than 0.2 s. Fig. 17 illustrates the battery charge/discharge signal, where the absence of the chattering mode is observed. The results obtained in this work improve the ones present in the literature. In [1], the short and long term strategy preserves the fuel cells but leads to an excessive use of the battery; the integrations of a minimum and maximum state of charge of the battery in [3] to help preserving the battery are also noticed.

However, the method used is a day-ahead scheduling, which does not consider the chattering mode. The method developed in this paper helps preserving the battery by taking into account the minimum and the maximum state of charge, the minimum and the maximum operating voltage and current, and it reduces chattering mode due to high frequency switching. On the other hand, the system performance is improved, static error e_S is kept inferior to 1% where it is inferior to 5% in [4]. Finally, the negotiations between the agents avoid the use of the battery to reduce the load voltage fluctuation. This improves the strategies used in [7]-[9].

A storage prioritization method is implemented by using the battery in priority to preserve the fuel cells or to maximize the hydrogen production.

VIII. Conclusion

In this paper, the isolated micro grid management is addressed. A hybrid approach has been developed; it is based on MAS for management and Simulink application as interface with the real process. MAS negotiation ensures an optimal decision making. Each resource agent gives a best deal according to data and constraints of its counterpart in the process, so the global decision is made based on real-time data with appropriate action for each device. The developed approach distributed guarantees modularity and speed of execution. The contribution of this paper is that unlike other approaches described in the literature, the developed method help preserving the battery storage system by considering chattering suppression, battery voltage, and current limits in addition to state of charge limits. Furthermore, the proposed method enhances the PV system power tracking performance presented in literature and, at the same time, controls load voltage. Moreover, results demonstrate a fast response time of the management system on solar and load variation as well. Future research work will be directed by considering economic criteria and electrical converter constraints in order to provide a global optimization solution.

Appendix

The data exchanged between Simulink and the MAS is formatted on the following format: *xxxyyyzzz*12345, where *xxxyyyzzz* is the variable id, *xxxyyy* is the device id/agent name, 12345 is the variable value, and *xxx* is the device type (bat for battery, src for photovoltaic source ...), *yyy* is the device number (a unique id per device type), *zzz* is the variable name. Agents on the MAS are named *xxxyyy* according to the device's representatives, except for the communication agents, i.e. a battery's state of charge of 32.15 % is transmitted "bat001soc32.15" to the MAS.

Acknowledgements

This work was supported by the laboratory of Simulation, Control, Analysis and Maintenance of Electrical Networks "SCAMRE" at the National Polytechnic School Maurice Audin of Oran, Algeria.

References

[1] G Bruni, S Cordiner, M Galeotti, V Mulone, Nobile M, Rocco V, Control strategy influence on the efficiency of a hybrid photovoltaic-battery-fuel cell system distributed generation system for domestic applications, *Energy Procedia* 2014, 45:237–46.

[2] Wahjono, E., Anggriawan, D., Sunarno, E., Nugraha, S., Tjahjono, A., Maximum Power Point Tracking of Photovoltaic Module for Battery Charging Based on Modified Particle Swarm Optimization, (2017) *International Review on Modelling and Simulations (IREMOS)*, 10 (1), pp. 77-84. doi: https://doi.org/10.15866/iremos.v10i1.11174

[3] Conte F, D'Agostino F, Pongiglione P, Saviozzi M, Silvestro F. Mixed-integer algorithm for optimal dispatch of integrated PVstorage systems. *IEEE Trans Ind Appl*, Vol. 55(Issue 1):238-247, 2019.

[4] Tian Y, Fan L, Tang Y, Wang K, Li G, Wang H. A coordinated multi-time scale robust scheduling framework for isolated power system with ESU under high RES penetration. *IEEE Access*, Vol 6:9774-9784, 2018.

[5] Farhat, M., Flah, A., Sbita, L., Photovoltaic Maximum Power Point Tracking Based on ANN Control, (2014) *International Review on Modelling and Simulations (IREMOS)*, 7 (3), pp. 474-480. doi: https://doi.org/10.15866/iremos.v7i3.1212

[6] E Dursun, O Kilic, Comparative evaluation of different power management strategies of a standalone PV/Wind/PEMFC hybrid power system, *Int J Electr Power Energy Syst*, Vol. 34(Issue 1):81-9, January 2012.

[7] F Zhang, K Thanapalan, A Procter, S Carr, J Maddy, G Premier, Power management control for off-grid solar hydrogen production and utilisation system, *Int J Hydrog Energy*, Vol. 38(Issue 11):4334-41, April 2013.

[8] M Trifkovic, WA Marvin, M Sheikhzadehy, P Daoutidis, Dynamic real-time optimization and control of a hybrid energy system, *European Control Conference (ECC)*, p. 2669–74, Zurich, July 2003.

[9] Amirullah, A., Penangsang, O., Soeprijanto, A., High Performance of Unified Power Quality Conditioner and Battery Energy Storage Supplied by Photovoltaic Using Artificial Intelligent Controller, (2018) *International Review on Modelling and Simulations (IREMOS)*, 11 (4), pp. 221-234. doi: https://doi.org/10.15866/iremos.v11i4.14742

Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved

[10] Attia, H., A Stand-Alone Solar PV System with MPPT Based on Fuzzy Logic Control for Direct Current Portable House Applications, (2018) *International Review on Modelling and Simulations (IREMOS)*, 11 (6), pp. 377-385. doi: https://doi.org/10.15866/iremos.v11i6.16074

[11] Attia, H., Artificial Neural Networks Based Maximum Power Point Tracking Photovoltaic System for Remote Park LED Lighting Applications, (2018) *International Review on Modelling and Simulations (IREMOS)*, 11 (6), pp. 396-405. doi: https://doi.org/10.15866/iremos.v11i6.16165

[12] C Bordons, F García-Torres, L Valverde, Optimal Energy Management In Microgrids With Renewable Generation, *Rev Iberoam Automática Inf Ind RIAI*, Vol. 12(Issue 2):117-32, April 2015.

[13] R Dufo-López, JL Bernal-Agustín, J Contreras, Optimization of control strategies for stand-alone renewable energy systems with hydrogen storage, *Renew Energy*, Vol. 32(Issue 7):1102-26, June 2007.

[14] S Abedi, A Alimardani, GB Gharehpetian, GH Riahy, SH Hosseinian, A comprehensive method for optimal power management and design of hybrid RES-based autonomous energy systems, *Renew Sustain Energy Rev*, Vol. 16(Issue 3):1577-87, April 2012.

[15] Hu W, Wang P, Gooi HB. Toward optimal energy management of microgrids via robust two-stage optimization. *IEEE Trans Smart Grid*, Vol 9(Issue 2):1161-1174, March 2018.

[16] Romero-Quete D, Cañizares CA. An affine arithmetic-based energy management system for isolated microgrids. *IEEE Trans Smart Grid*, Vol 10:2989-2998, May 2019.

[17] Paul, T.G.; Hossain, S.J.; Ghosh, S.; Mandal, P.; Kamalasadan, S. A Quadratic Programming Based Optimal Power and Battery Dispatch for Grid-Connected Microgrid. *IEEE Trans Ind. Appl*, Vol 54 (Issue 2):1793-1805, March 2018.

[18] Kumar, K.P.; Saravanan, B. Day ahead scheduling of generation and storage in a microgrid considering demand Side management. *Energy Storage*, Vol 21:78-86, February 2019.

[19] Hadidian MM, Kalam A, Nowdeh S, Ahmadi A, Babanezhad M, Saha S. Optimal sizing and energy management of stand-alone hybrid photovoltaic/wind system based on hydrogen storage considering LOEE and LOLE reliability indices using flower pollination algorithm. *Renew Energy*, Vol 135:1412–34, May 2019.

[20] Kozlov A, Tomin N, Sidorov D, Lora E, Kurbatsky V. Optimal operation control of PV-biomass gasifier-diesel-hybrid systems using reinforcement learning techniques. *Energies*, Vol 13(Issue 10):2632, May 2020.

[21] LM Fernandez-Ramírez, P Garcıa-Tribiño, AJ Gil-mena, F Llorens-iborra, F Jurado, CA García-V-azquez, Optimized operation combining costs, efficiency and lifetime of a hybrid renewable energy system with energy storage by battery and hydrogen in grid-connected applications, *Int J Hydrog Energy*, Vol. 41(Issue 48):23132-44, December 2016.

[22] P García, JP Torreglosa, LM Fernández, F Jurado, Optimal energy management system for stand-alone wind turbine/photovoltaic/hydrogen/battery hybrid system with supervisory control based on fuzzy logic, *Int J Hydrog Energy*, Vol. 38(Issue 33):14146-58, November 2013.

[23] JP Torreglosa, P García-triviño, LM Fernández-ramirez, F Jurado, Control based on techno-economic optimization of renewable hybrid energy system for stand-alone applications, *Exp Syst App*, Vol. 51:59-75, June 2016.

[24] L Valverde, F Pino, F Rosa, Definition, analysis and experimental investigation of operaion modes in hydrogen-renewable-based power plants incorporating hybrid energy storage, *Energy Convers Manag*, Vol. 113:290-311, April 2016.

[25] Hossain, M.A.; Pota, H.R.; Squartini, S.; Abdou, A.F. Modified PSO algorithm for real-time energy management in grid-connected microgrids. *Renew. Energy*, Vol 136:746-757, June 2019.

[26] Ahmad, J.; Imran, M.; Khalid, A.; Iqbal,W.; Ashraf, S.R.; Adnan, M.; Ali, S.F.; Khokhar, K.S. Techno economic analysis of a wind-photovoltaic-biomass hybrid renewable energy system for rural electrification: A case study of Kallar Kahar. *Energy*, Vol 148:208-234, April 2018.

[27] Firouzmakan P, Hooshmand R, Bornapour M, Khodabakhshian A. A comprehensive stochastic energy management system of micro-CHP units, renewable energy sources and storage systems in microgrids considering demand response programs. *Renew Sustain Energy Rev*, Vol 108:355–68, July 2019.

[28] EK Hussain, S Member, CM Bingham, D Stone, Hybrid stand-alone renewable energy system with high fuel-cell efficiency and unity power factor, *Proceedings of IEEE PES conference on innovative smart grid technologies*, pp. 1–6, Middle East (ISGT Middle East), December 2011.

[29] X Li, X Jiao, L Wang, Coordinated power control of wind-PV-fuel cell for hybrid distributed generation systems, *SICE annual conference (SICE)*, 2013, p. 150–5.

[30] JP Torreglosa, P García, LM Fernández, F Jurado, Energy dispatching based on predictive controller of an off-grid wind turbine/photovoltaic/hydrogen/battery hybrid system., *Renew Energy* 2015, 74:326–36.

[31] C Ziogou, D Ipsakis, C Elmasides, F Stergiopoulos, S Papadopoulou, P Seferlis, S Voutetakis, Automation infrastructure and operation control strategy in a standalone power system based on renewable energy sources, *J Power Sources*, Vol. 196(Issue 22): 9488–99, November 2011.

[32] P García-Triviño, F Llorens-Iborra, CA García-Vázquez, AJ Gil-Mena, LM FernándezRamírez, F Jurado, Long-term optimization based on PSO of a grid-connected renewable energy/battery/hydrogen hybrid system, *Int J Hydrog Energy*, Vol. 39(Issue 21):10805-16, July 2014.

[33] B Boudiaf, S Zebirate, N Aissani, A Chaker, *A Multi-agent Based Platform for Safety Control* (Springer, Cham, 2015, pp. 135-143).

Authors' information



Brahim Boudiaf received the Eng. degree in instrumentation engineering from the National Institute of Maintenance and Safety, Oran, Algeria in 2011. A Magister degree in electrical engineering from the National Polytechnique School of Oran, Oran, Algeria in 2014. He is currently a Ph.D. candidate in the National Polytechnique School of Oran, Algeria within "SCAMRE" Laboratory. His research interests include smart grids, fuzzy systems, renewable energy, demand response management, data driven load modeling and control, optimization theory and applications to smart grid issues.

Soraya Zebirate is a professor at the university of Oran 2, Algeria, department of Industrial Engineering. She obtained her doctorate degree and her qualification to direct research in automatic control at the university of Sidi Bel Abbes in Algeria. Team leader in "SCAMRE" Laboratory, her research area is focused on intelligent control of industrial systems.

Nassima Aissani is a Lecturer in Department Maintenance and Industrial Safety, Oran University 2, Mohamed Ben Ahmed, Algeria. She gets PhD degrees from both, University of Oran, Algeria and University of Valenciennes and Hainaut Cambrésis, France in the field of intelligent distributed control in industry. Her research area includes distributed control system, intelligent approaches and safety systems.



Abdelkader Chaker is a professor in the Department of Electrical Engineering at the Polytechnic national school of Oran in Algeria. He received a Ph.D degree in Engineering Systems from the University of Saint Petersburg. Director of "SCAMRE" Laboratory. His research activities include the control of large power systems, multimachine, multiconverter systems and the unified power flow controller. His teaching includes neural process control and real time simulation of power systems.

Copyright © 2021 Praise Worthy Prize S.r.l. - All rights reserved